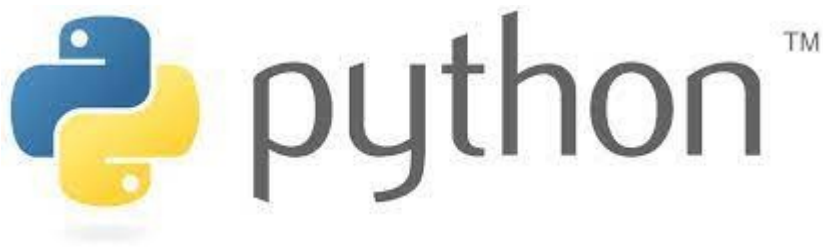


Introduction to Programming with Python



- ✓ Python is an interpreted high-level general-purpose programming language
- ✓ Python was conceived in the late 1980s by **Guido van Rossum**
- ✓ Its implementation was started in December 1989



Guido van Rossum

Why Learn?



- ✓ Python provides many useful features to the programmer.
- ✓ These features make it most popular and widely used language.
- ✓ few-essential feature of Python are.

1. Easy to use and Learn

2. Expressive Language

3. Interpreted Language

4. Object-Oriented Language

5. Open Source Language

6. Extensible

7. Wide Range of Libraries and Frameworks

8. GUI Programming Support

9. Embeddable

10. Dynamic Memory Allocation

11. Automatic garbage collection.



Why Learn?



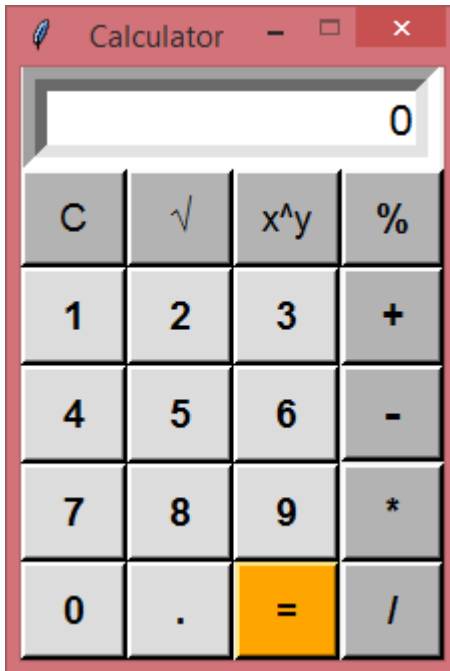
- Python works on different platforms (Windows, Mac, Linux, Raspberry Pi, etc).
- Python has a simple syntax similar to the English language.
- Python has syntax that allows developers to write programs with fewer lines than some other programming languages.
- Python code can be executed as soon as it is written.
- Python can be treated in a procedural way, an object-orientated way or a functional way



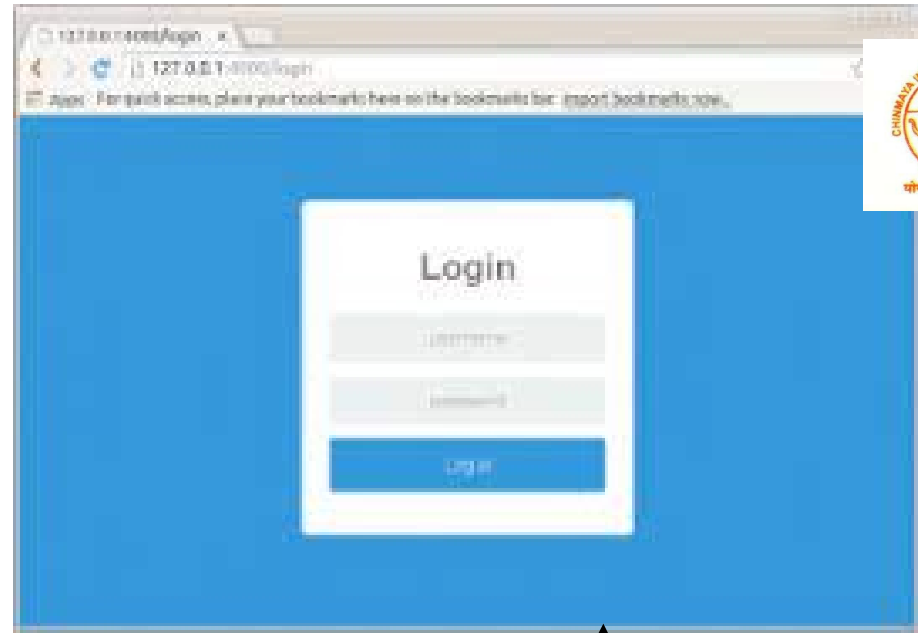
Where is Python used?



- Web Development. Python can be used to make web-applications at a rapid rate.
- Game Development.
- Machine Learning and Artificial Intelligence.
- Data Science and Data Visualization.
- Desktop GUI. [use PYQT, TKINTER]
- Web Scraping Applications.
- Business Applications.



GUI application ↑



web application ↑

game application →



Programming basics



- **Python is Interpreted** – Python is processed at runtime by the interpreter. You do not need to compile your program before executing it. This is similar to PERL and PHP.
- **Python is Interactive** – You can actually sit at a Python prompt and interact with the interpreter directly to write your programs.
- **Python is Object-Oriented** – Python supports Object-Oriented style or technique of programming that encapsulates code within objects.
- **Python is a Beginner's Language** – Python is a great language for the beginner-level programmers and supports the development of a wide range of applications from simple text processing to WWW browsers to games.

Python Features



Easy-to-learn – Python has few keywords, simple structure, and a clearly defined syntax. This allows the student to pick up the language quickly.

Easy-to-read – Python code is more clearly defined and visible to the eyes.

Easy-to-maintain – Python's source code is fairly easy-to-maintain.

A broad standard library – Python's bulk of the library is very portable and cross-platform compatible on UNIX, Windows, and Macintosh.

Interactive Mode – Python has support for an interactive mode which allows interactive testing and debugging of snippets of code.

Python Features



Portable – Python can run on a wide variety of hardware platforms and has the same interface on all platforms.

Extendable – You can add low-level modules to the Python interpreter. These modules enable programmers to add to or customize their tools to be more efficient.

Databases – Python provides interfaces to all major commercial databases.

GUI Programming – Python supports GUI applications that can be created and ported to many system calls, libraries and windows systems, such as Windows MFC, Macintosh, and the X Window system of Unix.

World-class companies that use python



Instagram

amazon



Uber



Netflix



Spotify



Facebook



Google



Dropbox



Pinterest



SMP

CHINTECH 10

First python program



- Example
- `print("Hello, World!")`

python install



- Many PCs and Macs will have python already installed.
- To check if you have python installed on a Linux or Mac, then on linux open the command line or on Mac open the Terminal and type:

➤ `$python --version`

python install



Step 1: Update and Refresh Repository Lists

Open a terminal window, and enter the following:

sudo apt update

Step 2: Install Supporting Software

The software-properties-common package gives you better control over your package manager by letting you add PPA

sudo apt install software-properties-common

Step 3: Add Deadsnakes PPA

Deadsnakes is a PPA with newer releases than the default Ubuntu repositories. Add the PPA by entering the following:

sudo add-apt-repository ppa:deadsnakes/ppa

Refresh the package lists again:

sudo apt update

Step 4: Install Python 3

Now you can start the installation of Python 3.8 with the command:

sudo apt install python3.8

Allow the process to complete and verify the Python version was installed successfully:

python --version

```
shajeer-mp@shajeer-mp:~$ python3 --version
Python 3.5.2
```

python install for Windows



Installing Python on Windows will take some important steps :

- Step 1): Go to the python official website at <https://www.python.org/downloads/windows/>.
- Step 2): Choose the latest version of Python releases for Windows.
- Step 3): After choosing the correct released version, Click on the download Python.
- Step 4): Click on Install now, and you can add **python.exe** path.

Applications



1. Data Science

Python is an entrusted language for scientific computing tasks, including analysis and visualization of large data sets.

2. Automation

Python allows developers to script custom automation and bring efficiency in tasks in less time. Automating repetitive tasks such as sending emails & voicemails, organizing files & folders, launching programs, filling out forms, etc. can be automated through scripting.

Applications



3. Application Development

Python is a favourable choice for web application development. Python integrates well with other programming languages and has some amazing web frameworks that boost the application development cycle.

4. AI & Machine Learning

Python's pre-built libraries are the reason why it is preferred for AI & machine learning development. **Numpy** for scientific computation, **Pybrain** for machine learning tasks, **Scipy** for technical and scientific computing are some of the python libraries that make it a supportive technology for AI and ML development.

Applications



5. Audio/Video Applications

Python is a sought-after programming language for creating audio/video applications. An example of this is the **Spotify** app that's built using Python programming language. Some of the python libraries such as PyDub, OpenCV support building audio/video streaming apps with high-level functionality.

6. Console Applications

A console application is a computer program that's designed for the text-only interfaces, such as the command-line interface of Unix, DOS operating system.

Applications



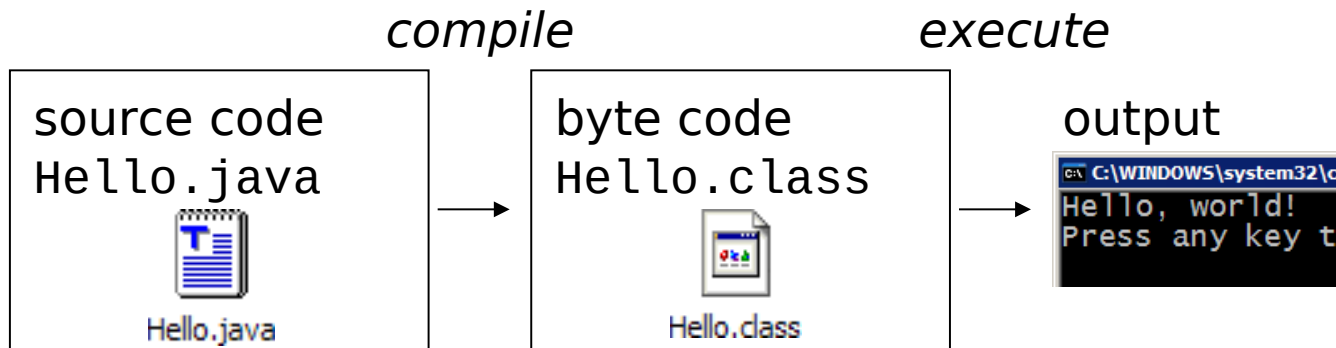
7. Desktop GUI

Python has libraries for building desktop GUI applications as well. Camelot, Kivy, PyGTK are some of the amazing libraries and tool-kits available for building desktop GUI.

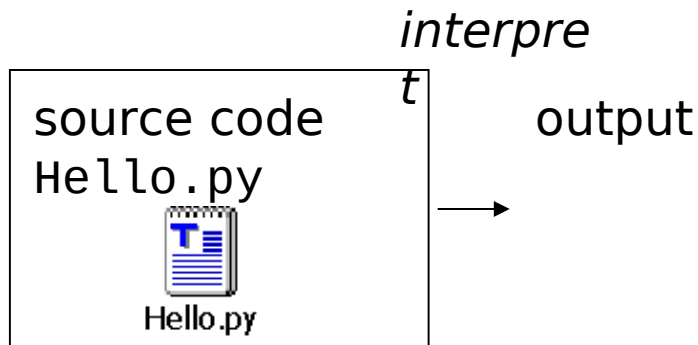
Compiling and interpreting



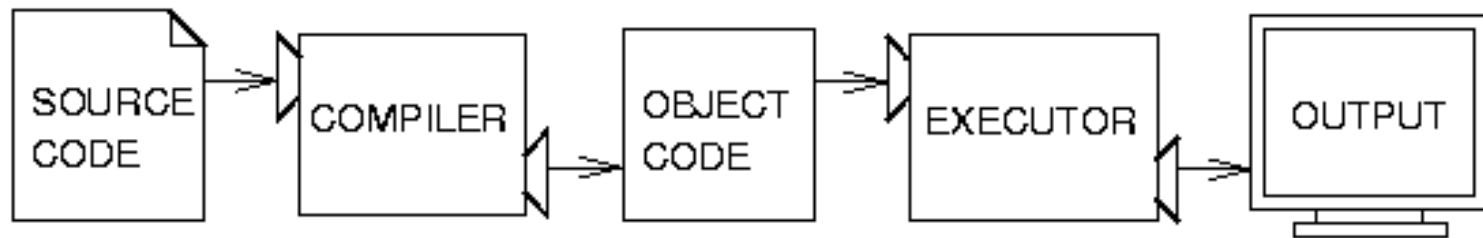
- Many languages require you to *compile* (translate) your program into a form that the machine understands.



- Python is instead directly *interpreted* into machine instructions.



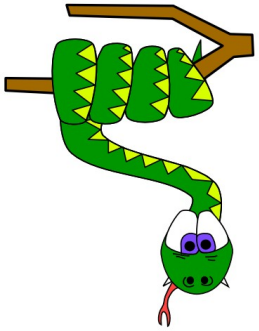
Compiling and interpreting



python Quick start



- Python is an interpreted programming language, this means that as a developer you write Python (.py) files in a text editor and then put those files into the python interpreter to be executed.
- `$ python helloworld.py`
- Where "helloworld.py" is the name of your python file.



Different Python Programming Environment

Atom



1. Atom : Atom is a free and open-source text and source code editor for macOS, Linux, and Microsoft Windows with support for plug-ins written in Node.js, and embedded Git Control, developed by GitHub. Atom is a desktop application built using web technologies.



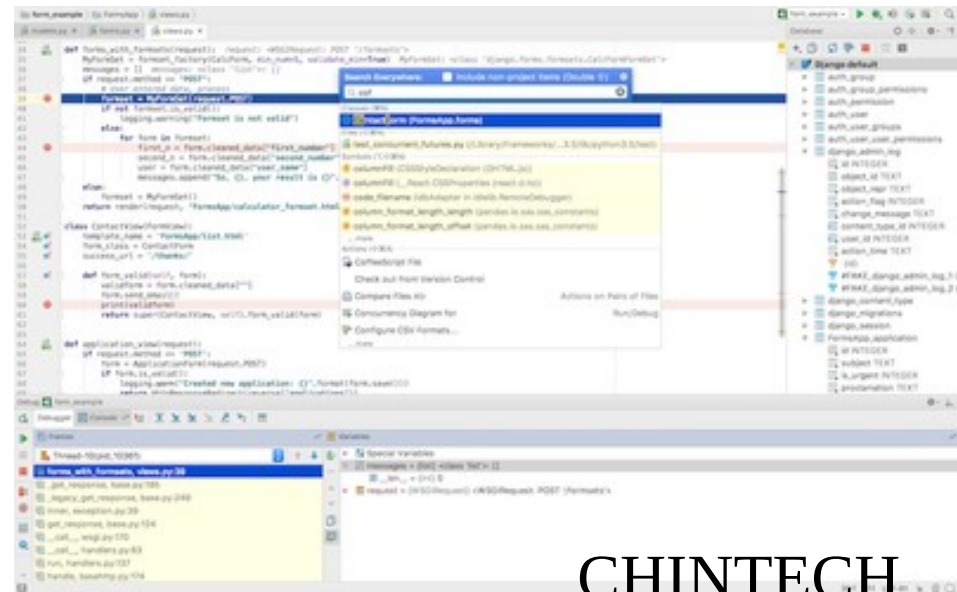
PyCharm



- PyCharm is an integrated development environment (IDE) designed for programmers, to provide all the tools you need for productive Python development.
- It is developed by the Czech company JetBrains. It provides code analysis, a graphical debugger, an integrated unit tester, integration with version control systems (VCSes), and supports web development with Django as well as data science with Anaconda



SMP



CHINTECH 5

The Jupyter Notebook



The Jupyter Notebook is an open-source web application that allows you to create and share documents that contain live code, equations, visualizations and narrative text.

Uses:

- Programming in the browser
- Code, instructions and output are displayed “in-line”
- Used by scientists and researchers

<https://jupyter.org/>



Visual Studio Code



Visual Studio Code is a free source-code editor made by Microsoft for Windows, Linux and macOS. Features include support for debugging, syntax highlighting, intelligent code completion, snippets, code refactoring, and embedded Git.

[link - <https://code.visualstudio.com/>]



Spyder



Spyder : Spyder is an open-source cross-platform integrated development environment for scientific programming in the Python language. [link - <https://www.spyder-ide.org/>]



Google Colab



As a programmer, you can perform the following using Google Colab.



Write and execute code in Python

Document your code that supports mathematical equations

Create/Upload/Share notebooks

Import/Save notebooks from/to Google Drive

Import/Publish notebooks from GitHub

Import external datasets e.g. from Kaggle

Integrate PyTorch, TensorFlow, Keras, OpenCV

Free Cloud service with free GPU